MIL-Q-12241B(Ord)
16 September 1959
SUPERSEDING
MIL-Q-12241A(Ord)
27 June 1958

MILITARY SPECIFICATION
QUADRANTS, ELEVATION, M13 AND M13A1; PACKAGING OF

1. SCOPE

1.1 This specification covers the packaging and packing of Elevation Quadrant M13, M13A1, with or without Instrument Light T22, for each level of protection.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant

Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed, Flexible (Waterproofed)
MIL-B-130	Barrier, Material, Paper, Noncorrosive
MIL-L-3150	Oil, Preservative, Medium
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and methods.- The materials, methods and procedures used in the packaging of items shall be in accordance with this specification. Materials and methods not covered by a specification shall be suited to the intended use and subject to the approval of the contracting officer.

3.1.1 Materials.- The following shall apply wherever packaging materials are referred to in this specification by name only:

REFERENCE	FULL IDENTIFICATION
Flexible single face corrugated fiberboard	Type I, class 1 of Specification PPP-P-291
Greaseproofed barrier material	Type I, grade A, class 2 of Specification MIL-B-121
Gummed tape	Class 2 of Specification UU-T-111
Noncorrosive barrier material	Type II of Specification MIL-B-130
Pressure-sensitive tape	Type I or II of Specification UU-T-106
Water-vaporproof bag	Type I, class e of Specification MIL-B-117

3.2 Size of containers.- Bags, interior containers and exterior containers shall be of a minimum size that will adequately contain the items.

3.3 Preservation and packaging.

3.3.1 Level A.

3.3.1.1 Cleaning.- The exterior surfaces of each quadrant and light shall be cleaned so as to be free from dirt, smudges, fingerprints and other foreign substances in accordance with Process C-1 of Specification MIL-P-116.

3.3.1.2 Drying.- Immediately after cleaning, all items shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

3.3.1.3 Preservative application.- The mounting hardware for the quadrant and light shall be coated with preservative conforming to Specification MIL-L-3150 (P-7 of Specification MIL-P-116). The preservative shall be applied on the hardware by any applicable procedure specified in Specification MIL-P-116. Excess preservative shall be removed.

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3.3.1.4 Unit packaging.

3.3.1.4.1 Quadrant without the instrument light.- The quadrant shall be packaged as specified herein, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.4.1.1 Quadrant mounting hardware.- The preservative coated mounting hardware for the quadrant shall be wrapped with greaseproofed barrier material, which shall be secured in place with pressure-sensitive tape. The wrapped hardware shall be secured to the housing with pressure-sensitive tape.

3.3.1.4.1.2 Level vial cover.- The metal cover on the level vial assembly shall be positioned so that the face of the level vial is covered. The metal cover shall be secured in place with pressure-sensitive tape.

3.3.1.4.1.3 Reflector.- The reflector shall be covered with at least four thicknesses of tissue paper which conforms to Specification MIL-P-13988. The tissue paper shall be secured in place with pressure-sensitive tape.

3.3.1.4.1.4 Wrapping the quadrant.- The quadrant shall be wrapped with flexible single face corrugated fiberboard. The wrap shall be secured in place with gummed tape.

3.3.1.4.1.5 Bagging the wrapped quadrant.- The wrapped quadrant shall be placed in a water-vaporproof bag. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

3.3.1.4.1.6 Fiberboard box.- The bagged quadrant shall be immobilized in a fiberboard box conforming to class 2 or 3, any compliance symbol and any style of Specification PPP-B-636. The box shall be closed with pressure-sensitive tape conforming to Specification PPP-T-76 in accordance with the appendix of Specification PPP-B-636.

3.3.1.4.2 Quadrant with instrument light.- The quadrant and instrument light shall be packaged as specified herein and as shown in Figure 1, which is in conformance with Method IA-8 of Specification MIL-P-116.

3.3.1.4.2.1 Instrument light mounting hardware.- The preservative coated mounting hardware for the instrument light shall be wrapped with greaseproofed barrier material, which shall be secured in place with pressure-sensitive tape. The wrapped hardware shall be secured to the instrument light housing with pressure-sensitive tape.

3.3.1.4.2.2 Wrapping the light.- The instrument light shall be wrapped with flexible single face corrugated fiberboard. The wrap shall be secured in place with gummed tape.

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3.3.1.4.2.3 Bagging the wrapped light.- The wrapped light shall be placed in a water-vaporproof bag. The bag shall be heat sealed in such a manner as to pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

3.3.1.4.2.4 Quadrant.- The quadrant shall be packaged as specified in 3.3.1.4.1.1 to 3.3.1.4.1.5 inclusive.

3.3.1.4.2.5 Fiberboard box.- The bagged instrument light and bagged quadrant shall be immobilized in a fiberboard box conforming to the characteristics (except for size) of the box specified in 3.3.1.4.1.6. The box shall be closed as specified in 3.3.1.4.1.6.

3.3.2 Level C.

3.3.2.1 Cleaning and drying.- Cleaning and drying shall be as specified in 3.3.1.1 and 3.3.1.2.

3.3.2.2 Unit packaging.- Packaging shall be as specified herein, which is in conformance with Method III of Specification MIL-P-116.

3.3.2.2.1 Quadrant without the instrument light.- Packaging shall be as specified in 3.3.1.4.1.1 to 3.3.1.4.1.6 inclusive, except as follows:

(a) Since the preservative is not required, the mounting hardware shall be wrapped with noncorrosive barrier material in lieu of the greaseproofed barrier material specified in 3.3.1.4.1.1.

(b) The water-vaporproof bag, specified in 3.3.1.4.1.5, shall be omitted.

(c) A fiberboard box, conforming to type I or II, class 1, any style of Specification PPP-B-636, shall be used in lieu of the fiberboard box specified in 3.3.1.4.1.6. The box shall be closed as specified in the appendix to the box specification.

3.3.2.2.2 Quadrant with instrument light.- Packaging shall be as specified in 3.3.1.4.2.1 to 3.3.1.4.2.5 inclusive, except as follows:

(a) Since preservative is not required, the mounting hardware shall be wrapped with noncorrosive barrier material in lieu of the greaseproofed barrier material specified in 3.3.1.4.2.1.

(b) The water-vaporproof bag, specified in 3.3.1.4.2.3, shall be omitted.

(c) A fiberboard box conforming to the characteristics (except for

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size) of the fiberboard box, specified in 3.3.2.2.1 c, shall be used in lieu of the box specified in 3.3.1.4.2.5. The box shall be closed as specified in 3.3.2.2.1 c.

3.4 Packing.

3.4.1 Level A.- Sixty unit packaged quadrants without instrument lights or forty unit packaged quadrants with packaged instrument lights shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-601 - Style A or B (oversea type) - type II or III, class 2 plywood
PPP-B-621 - Class 2 (oversea), style 2 or 4

3.4.2 Level B.- Sixty unit packaged quadrants without instrument lights or forty unit packaged quadrants with packaged instrument lights shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1 (domestic) style 2 or 4

3.4.3 Level C.- A number of unit packaged quadrants, with or without packaged instrument lights, shall be immobilized in a shipping container conforming to one of the following specifications:

PPP-B-585 - Any style except style 1, class 1, construct for type 3 load
PPP-B-601 - Any style under domestic type
PPP-B-621 - Class 1, (domestic), any style
PPP-B-636 - Type I or II, class I, any style, except FTC, TS, OPF and FPF

3.4.3.1 Gross weight.- The gross weight for a Level C shipping container shall not exceed 200 pounds for a wood container and the weight limits specified in Specification PPP-B-636 shall not be exceeded for a fiberboard container.

3.4.4 Strapping.- Strapping shall be applied on shipping containers as specified in the applicable container specification.

3.5 Marking.

3.5.1 Interior packages.- Each water-vaporproof bag and each box shall be marked in accordance with Standard MIL-STD-129. The serial

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number shall be added in the identification data on each Level A unit package and omitted on the Level C unit package.

3.5.2 Shipping containers.- Each shipping container shall be marked in accordance with Standard MIL-STD-129. If the serial numbers of the quadrants in a Level A or B shipping container are in consecutive order, the serial numbers shall be indicated as specified in Standard MIL-STD-129. If the serial numbers are not in consecutive order, they shall be indicated on packing lists.

3.5.3 Stock number.- The stock number (or numbers if there is a light) to be added in the identification data shall be obtained from the contracting officer.

3.6 Preproduction pack.- Prior to quantity production, a preproduction pack, consisting of a complete and packed shipping container, shall be inspected to determine conformance with the applicable requirements of this specification. A rough handling test is required.

3.7 Workmanship.- All operations involved in accomplishing corrosion prevention, preservation, packaging, and packing by the procedures specified herein shall be in accordance with the highest grades of practice associated with this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

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4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of the detail specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by the detail specification of other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Inspection provisions.- Quality assurance provisions specified in Specification MIL-P-116 shall apply where applicable and are not in conflict with the provisions specified herein.

4.4.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.4.2 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.3 Examination.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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4.4.3.1 Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL values are expressed in percent defective

MAJOR AQL 2.5	Req. Para.	Test Method
1. Marking illegible, incorrect, missing, not conforming to specifications (Levels A, B, C)	3.5	Visual
2. Improper cleaning (Levels A, C)	3.3.1.1	MIL-P-116-Visual
3. Improper drying (Levels A, C)	3.3.1.2	MIL-P-116-Visual
4. Preservative missing, inadequate (Level A)	3.3.1.3	MIL-P-116-Visual
5. Level vial not covered (Levels A, C)	3.3.1.4.1.2	Visual
6. Water-vaporproof bag damaged, improperly sealed (Level A)	3.3.1.4.1.5	Visual
7. Assemblies not immobilized (Levels A, C)	3.3.1.4.1.6 3.3.1.4.2.5	Visual
8. Barrier material missing, inadequate (Levels A, C)	3.3.1.4.2.1	Visual
9. Fiberboard missing, inadequate (Levels A, C)	3.3.1.4.2.2	Visual
10. Loose hardware (Levels A, C)	3.3.1.4.2.1	Visual
11. Minimum air void (Level A)	3.3.1.4.1.5 3.3.1.4.2.3	Visual-Tactile
MINOR AQL 4.0		
41. Tape missing, improperly applied (Levels A, C)	3.3.1.4.1.6	PPP-B-636-Visual
42. Reflector tissue missing, inadequate (Levels A, C)	3.3.1.4.1.3	Visual
43. Strapping missing, improperly applied (Levels A, B, C)	3.4.4	PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636-Visual

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TABLE I - CLASSIFICATION OF DEFECTS Cont'd

MINOR AQL 4.0	Req. Para.	Test Method
44. Shipping container not conforming to Specifications (Levels A, B, C)	3.4.1; 3.4.2; 3.4.3	PPP-B-585, PPP-B-601, PPP-B-621, PPP-B-636- Standard Measuring Equipment (SME)
45. Excessive gross weight (Level C)	3.4.3.1	SME
46. Workmanship (Levels A, B, C)	3.7	Visual

TABLE II - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105
For Reduced Inspection use Level L5, and Sampling Plan Table IV-A of Standard MIL-STD-105
AQL values are expressed in percent defective

MAJOR AQL 6.5	Req. Para.	Test Method
1. Heat seal (Level A)	3.3.1.4.1.5, 3.3.1.4.2.3	MIL-P-116
2. Quick leak (Level A)	3.3.1.4.1.5, 3.3.1.4.2.3	MIL-P-116
3. Preservative compound application (Level A)	3.3.1.3	MIL-P-116

4.5 Test methods and procedures.

4.5.1 Preproduction pack.- A preproduction pack shall be inspected by the contractor or his agent, and witnessed by the Government inspector to determine conformance with the requirements of this specification for the packaging and packing levels specified.

4.5.1.1 Preproduction pack failure.- Should the preproduction pack fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the process used to produce the preproduction pack. When corrections in the process have been made, a new preproduction pack shall be completely inspected by the contractor and witnessed by the Government inspector.

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4.5.2 Sample packs.- Sample packs for heat seal and quick leak tests shall be selected from acceptable packs of the sample that has met the minimum requirements listed in Table I.

4.5.3 Cleanliness, rough handling, heat seal, quick leak and preservative retention tests.- Cleanliness, rough handling, heat seal, quick leak and preservative retention tests shall be conducted as specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of packaging and packing

6.2 Size and weight of packed shipping container.

6.2.1 Quadrant without instrument light.- The following inside dimensions, overall cubage and gross weight are approximate for sixty unit packaged quadrants packed in a nailed wood box for Level A:

Inside dimensions	- 30. inches long x 17-1/4 inches wide x 10-1/2 inches deep
Cubage	- 4.2 cubic feet
Gross weight	- 145. pounds

6.2.2 Quadrant with instrument light.- The following inside dimensions, overall cubage and gross weight are approximate for forty unit packaged quadrants with packaged instrument lights packed in a nailed wood box for Level A:

Inside dimensions	- 23-3/4 inches long x 23 inches wide x 10 inches deep
Cubage	- 4.3 cubic feet
Gross weight	- 150. pounds

6.3 Definitions.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under

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procurement. This category also includes commercial testing equipment.

6.3.2 Tactile.- Skillful manipulation or sensing.

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QUADRANT, ELEVATION M13 - Packaging the quandrant, instrument light and mounting hardware.

FIGURE 1

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